

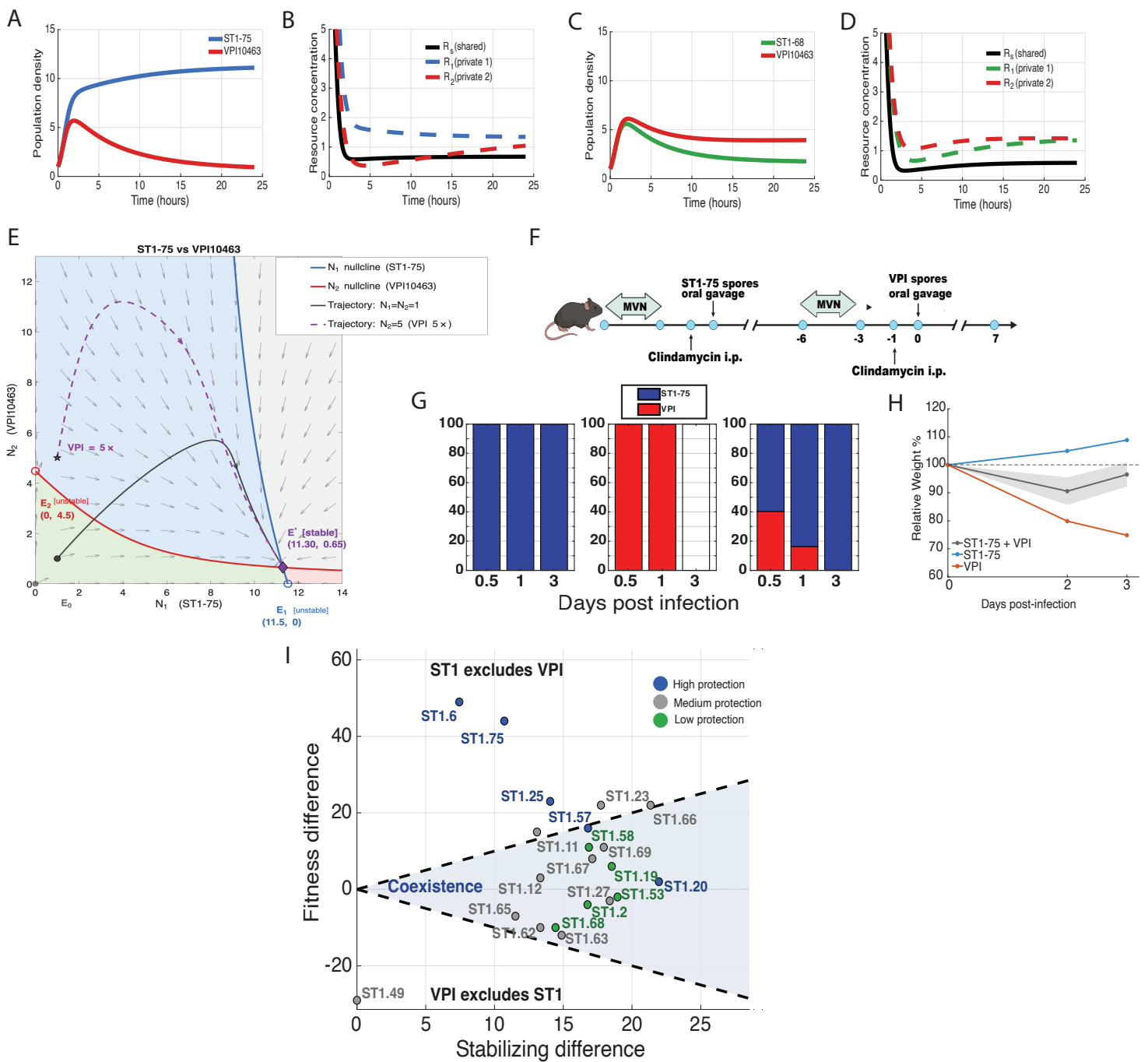


Figure 7: (A–B) Consumer resource model of competition between ST1.75 and VPI10463 strains grown on shared and private resources. (A) Population density dynamics over time (hours) for each strain. (B) Corresponding resource concentration dynamics, showing shared resource (R_0) and private resources (R_1 and R_2) for each strain. (C) Population density over time for each strain ST1.68 and VPI10463. (D) Resource concentration dynamics for shared and private resources, as in (B). (E) Phase-plane analysis of competition between ST1.75 and VPI10463. Solid blue and red lines indicate the N_1 (ST1.75) and N_2 (VPI10463) nullclines, respectively. Background shading indicates regions where only ST1.75 grows (blue), only VPI10463 grows (red), both strains grow (green), or neither grows (gray). Gray arrows indicate the vector field. Solid dark gray and dashed purple lines show representative trajectories initiated at $N_1 = N_2 = 1$ and $N_1 = 1, N_2 = 5$ (VPI 5 $\times$), respectively. Open circles mark the single-strain equilibria E_1 (ST1.75 alone; unstable) and E_2 (VPI10463 alone; unstable). The filled purple diamond marks the stable coexistence equilibrium E^* with coordinates indicated. (F) Experimental timeline for mouse infection model. Mice were pretreated with clindamycin (i.p.), followed by oral gavage of ST1-75 spores, then a second clindamycin dose. MVN (mock vehicle or treatment) was administered at indicated timepoints. VPI spores were administered by oral gavage at day -1 relative to the final timepoint. (G) Stacked bar plots showing the proportion of mice colonized by ST1.75 (blue) vs. VPI (red) at 0.5, 1, and 3 days post-infection across three experimental conditions (left to right panels). (H) Relative weight (%) of mice colonized with ST1.75 + VPI (blue) vs. VPI alone (orange) over 3 days post-infection. Shaded region indicates the normal weight range. (I) Ecological coexistence analysis of multiple ST1 isolates plotted by stabilizing difference (x-axis) vs. fitness difference (y-axis). Dots are colored by protection level (dark blue = high, gray = medium, green = low). Regions of ST1 exclusion of VPI, coexistence, and VPI exclusion of ST1 are indicated. Dashed lines demarcate predicted coexistence boundaries based on modern coexistence theory.